Spectral Quantum Memory for Implicit Neural Representations

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Code

NEURAL INFORMATION PROCESSING SYSTEMS

Why Spectral Memory

Implicit neural representations (INRs) model signals as continuous functions from coordinates to values, and their performance depends on how spectral capacity is allocated. Data reuploading circuits compute finite Fourier series. Encoding gates fix the frequencies and trainable parameters set the coefficients.

In existing multiband QINRs the bands operate largely independently. What one band computes reaches the others only through the classical decoder.

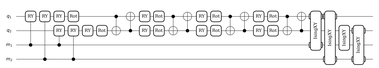
Key idea

Sequential data reuploading bands share a coherent quantum memory register. Each band reads the memory, computes on its own coordinate upload, and writes part of its state back, so later bands build on the computation of earlier ones.

Contributions

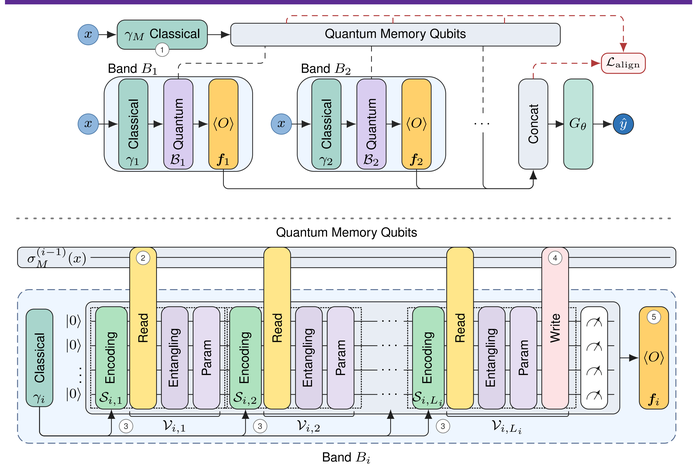
- **SQMM**, a memory coupled QINR that propagates coherent spectral context across data reuploading bands.
- **Theory.** The memory state stays an operator valued Fourier series supported on the Minkowski sum of the band spectra.
- **Spectral memory alignment loss** that trains memory to summarize the deployed prefix spectrum.
- **Results.** Best MSE among the evaluated compact INR and QINR baselines on audio and images.

One Band as Deployed



Band 3 of the trained Cello model. Controlled R_Y reads from memory qubits m_1, m_2. R_Y uploads with entangling layers on data qubits q_1, q_2, and XY writes to memory.

SQMM Architecture



Sequential quantum bands share the memory register M and emit Pauli expectation features, which a classical decoder combines. Dashed red arrows mark the training only alignment loss.

1. The memory is prepared from the coordinate by a one time upload $\gamma_M(x)$.
2. Each band reads the memory through input independent couplings.
3. It uploads its own coordinate encoding $\gamma_i(x) = A_i \mathrm{PE}(x) + b_i$ several times.
4. It writes part of its local state back into memory, and its local qubits are discarded.
5. It emits Pauli expectation features, and the updated memory passes to the next band. The memory is never measured.

Band i

$$\sigma_M^{(i)}(x) = \mathrm{Tr}_{D_i}\Big[\mathcal{B}_i(\gamma_i(x))\big(\eta_{D_i} \otimes \sigma_M^{(i-1)}(x)\big)\Big],$$
$$f_{i,a}(x) = \mathrm{Tr}\big[(O_{i,a} \otimes I_M)\,\xi_i(x)\big], \qquad \hat{y}(x) = G_\theta\big(\boldsymbol{f}_1(x), \ldots, \boldsymbol{f}_n(x)\big).$$

Memory Keeps the Fourier Structure

Every frequency that leaves a band, in its features or in the memory passed on, is the sum of a frequency already in the incoming memory and a frequency introduced by the band's own uploads. Input independent layers only set the coefficients.

Lemma (spectral propagation through memory)

If $\sigma_M^{(i-1)}(x) = \sum_\mu e^{\mathrm{i}\langle \mu, x\rangle} \tau_\mu^{(i-1)}$ and $\mathcal{B}_i(\gamma_i(x)) = \sum_{\nu \in \Omega_{i,x}^{\mathrm{loc}}} e^{\mathrm{i}\langle \nu, x\rangle} \mathcal{B}_{i,\nu}$, then

$$\Omega_{M,x}^{(i)} \subseteq \Omega_{M,x}^{(i-1)} + \Omega_{i,x}^{\mathrm{loc}}, \qquad \Omega_{f_i,x} \subseteq \Omega_{M,x}^{(i-1)} + \Omega_{i,x}^{\mathrm{loc}},$$
$$\tau_\omega^{(i)} = \sum_{\mu + \nu = \omega} \mathrm{Tr}_{D_i}\Big[\mathcal{B}_{i,\nu}\big(\eta_{D_i} \otimes \tau_\mu^{(i-1)}\big)\Big].$$

Memory creates no new primitive frequencies. After band i its support is contained in $\Omega_{M,x}^{(0)} + \sum_{j \le i} \Omega_{j,x}^{\mathrm{loc}}$, with matrix coefficients $\tau_\omega \in \mathbb{C}^{2^m \times 2^m}$ that later bands reuse and reweight.

Spectral Memory Alignment Loss

The lemma says memory can carry a spectral summary of the prefix computation. The alignment loss makes this summary readable.

Prefix feature spectrum. FFT coefficients $\bar{\boldsymbol{c}}_{j,h}^F$ of the features of band j at grid frequency $\bar{\omega}_h$,

$$P_i^F(\bar{\omega}_h) = \sum_{j=1}^{i} \big|\boldsymbol{c}_{j,h}^F\big|_2^2.$$

Memory spectrum. Probe memory with all 4^m Pauli strings Q_s,

$$P_i^M(\bar{\omega}_h) = 2^{-m} \sum_{Q_s} \big|\bar{c}_{i,s,h}^M\big|^2 \;\approx\; \big|\tau_{\bar{\omega}_h}^{(i)}\big|_F^2.$$

Objective. With normalization $\mathcal{N}_\delta$ and stop gradient,

$$\mathcal{L}_{\text{align}} = \sum_{i=1}^{n-1} \frac{1}{h} \Big(\mathcal{N}_\delta P_i^M(\bar{\omega}_h) - \mathrm{sg}\big[\mathcal{N}_\delta P_i^F(\bar{\omega}_h)\big]\Big)^2,$$
$$\mathcal{L} = \mathcal{L}_{\text{task}} + \lambda_{\text{align}}\, \mathcal{L}_{\text{align}}.$$

The auxiliary term trains the memory after each prefix to store the resolved spectrum of that prefix. Downstream bands can still reuse, refine, or correct upstream frequencies.

Does the Memory Matter?

Mechanism ablation on Cello, MSE $(\times 10^{-3})$. The control gives every band two further local qubits that are not shared, matching the qubit count and parameter budget.

Setting	Params	MSE↓ $(\times 10^{-3})$
No memory, local qubits	650	0.53±0.06
Memory	642	0.52±0.08
$\mathcal{L}_{\text{out}}$	642	0.54±0.06
$\mathcal{L}_{\text{align}}$ (Ours)	642	**0.48**±0.08

The no memory control has a higher error than SQMM. Directly supervising the output spectrum with $\mathcal{L}_{\text{out}}$ gives a similar error, while aligning the memory spectrum gives the best result at the same parameter count. Alignment also improves MSE, PSNR and LSD on Cello and MSE, PSNR, SSIM and LPIPS on every image.

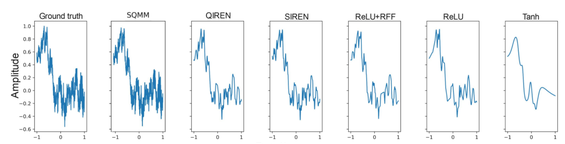
Takeaways

- **Fourier structure is kept.** Input independent memory read and write layers preserve the finite Fourier structure of data reuploading circuits.
- **Spectral context.** Later bands condition their coefficients on an operator valued spectral summary of earlier bands.
- **Memory as an interface.** Memory should be treated as a trainable spectral interface rather than an opaque hidden state.
- **Limitations.** All results are simulated. Simulation cost and hardware measurement overhead are the main limitations.

Results

Base benchmark (QIREN protocol), MSE $(\times 10^{-3})$. Subscript 2048 means expectations are estimated from 2048 shots.

	SQMM*	SQMM*$_{2048}$	QIREN	QVF	IGA	FINER	SIREN	MSA-INR	CAFE (enc.)
Params	642	642	649		673	691	673	673	703
Cello	**0.48**±0.08	0.76±0.06	6.79±0.35		4.87±0.01	3.73±0.44	5.02±0.08	7.04±1.23	3.60±0.60
Params	654	654	657	671	687	697	687	697	709
Astronaut	**1.12**±0.10	2.86±0.53	5.90±0.49	12.61±1.65	2.01±0.53	2.80±0.31	2.15±0.22	2.03±0.28	10.46±5.92
Camera	**0.38**±0.08	1.27±0.16	2.19±0.31	4.43±1.38	1.07±0.46	1.96±0.62	0.76±0.47	1.14±0.14	3.36±1.41
Coffee	**0.78**±0.05	1.79±0.16	2.52±0.67	14.38±6.80	1.63±0.42	1.85±0.31	2.26±0.40	1.37±0.19	5.17±1.06



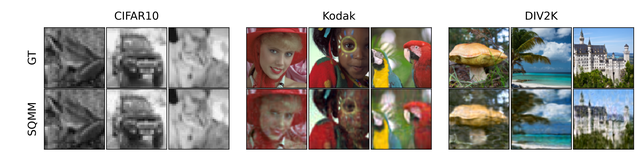
Cello reconstruction adapted from QIREN, normalized amplitude over normalized time.

Audio, GTZAN music, UrbanSound8K environmental sound and VCTK speech with 10 clips per class, MSE $(\times 10^{-3})$.

	SQMM*	QIREN	FINER	SIREN	NeRF	MSA-INR	IGA	CAFE (enc.)
Params	642	649	691	673	698	673	673	703
GTZAN	**4.71**±0.17	44.19±1.88	5.14±1.06	8.65±0.61	8.89±0.44	42.11±3.39	8.45±0.56	6.42±1.34
US8K	**6.11**±0.27	51.83±0.86	8.56±2.35	15.55±1.49	13.96±0.72	52.21±1.55	15.47±1.22	10.83±3.75
VCTK	**1.00**±0.03	10.70±0.10	1.66±0.21	2.98±0.15	2.59±0.15	8.51±0.90	2.69±0.13	2.25±0.34

Grayscale CIFAR10, 100 images.

	SQMM*	QIREN+	QVF	FINER	SIREN	IGA	MSA-INR
Params	654	801	671	697	687	687	697
SSIM↑ $(\times 10^{-2})$	**99.21**±0.03	97.69±0.19	92.90±0.23	97.43±0.39	98.37±0.32	98.56±0.27	98.59±0.14
LPIPS↓ $(\times 10^{-4})$	**3.44**±0.25	30.41±2.98	104.84±8.38	14.17±3.69	8.22±1.89	7.64±1.70	10.56±1.93
MSE↓ $(\times 10^{-3})$	**0.46**±0.02	1.67±0.13	5.12±0.17	1.45±0.21	0.98±0.21	0.98±0.18	0.90±0.11



Ground truth (top) and SQMM (bottom). CIFAR10 at 32×32, Kodak and DIV2K at 128×128.

Kodak and DIV2K in RGB from 32^2 to native size, MSE $(\times 10^{-3})$. NeRF holds $\times 1.54$ the parameters of SQMM.

	SQMM*	NeRF	MSA-INR	SIREN	IGA	R+RFF	FLAIR
Params	666	1023	747	717	717	723	723
Kodak 32^2	**1.58**±0.10	3.80±0.42	1.85±0.06	2.70±0.33	3.11±0.34	3.35±0.07	3.48±0.08
Kodak 64^2	**5.43**±0.03	8.76±0.68	6.84±0.06	6.89±0.26	7.06±0.25	9.47±0.18	9.57±0.46
Kodak 128^2	**11.54**±0.05	14.94±0.26	14.02±0.05	13.18±0.47	13.65±0.36	16.89±0.29	17.57±0.28
Kodak 256^2	**18.52**±0.04	21.57±0.23	21.54±0.04	20.57±0.55	20.94±0.54	24.53±0.34	25.37±0.14
Kodak	**32.08**±0.15	34.79±0.40	35.64±0.26	34.44±0.61	34.04±0.56	38.54±0.64	39.04±0.39
DIV2K 32^2	**3.13**±0.08	6.91±0.20	3.58±0.09	5.22±0.74	5.34±0.55	6.12±0.22	7.06±0.13
DIV2K 64^2	**11.13**±0.06	18.57±0.66	13.66±0.11	14.37±0.75	15.22±0.60	18.42±0.41	19.83±0.28
DIV2K 128^2	**23.16**±0.34	30.87±0.38	27.35±0.41	27.07±1.02	28.11±0.98	32.63±0.76	35.18±0.30
DIV2K 256^2	**35.15**±0.43	42.68±1.26	39.84±0.31	39.34±0.74	40.02±0.40	45.80±0.94	48.18±0.70
DIV2K 512^2	**54.13**±0.44	61.08±0.53	59.10±0.46	58.41±1.14	58.40±0.39	65.14±0.85	68.11±1.31